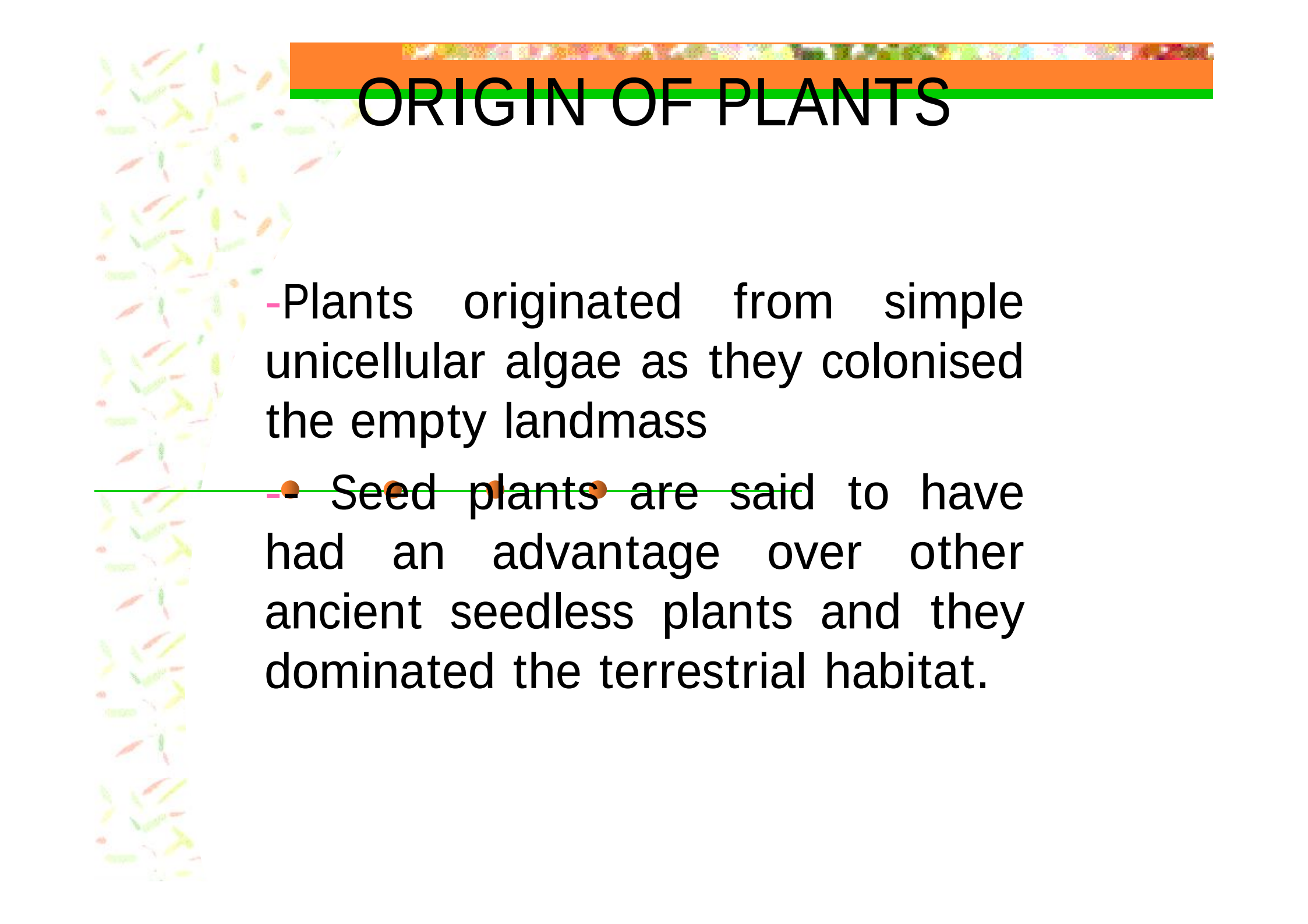




ORIGIN OF PLANTS

- Plants originated from simple unicellular algae as they colonised the empty landmass
- Seed plants are said to have had an advantage over other ancient seedless plants and they dominated the terrestrial habitat.



WHY PLANTS ARE IMPORTANT

- Plants are the backbone of all life on earth and a very essential resource for human well being
- Everyday life depends on plants:
- - **Food** – approx. 7000 species of different plants are used as food by humans
- - **Water** – the water cycle in nature is regulated by plants

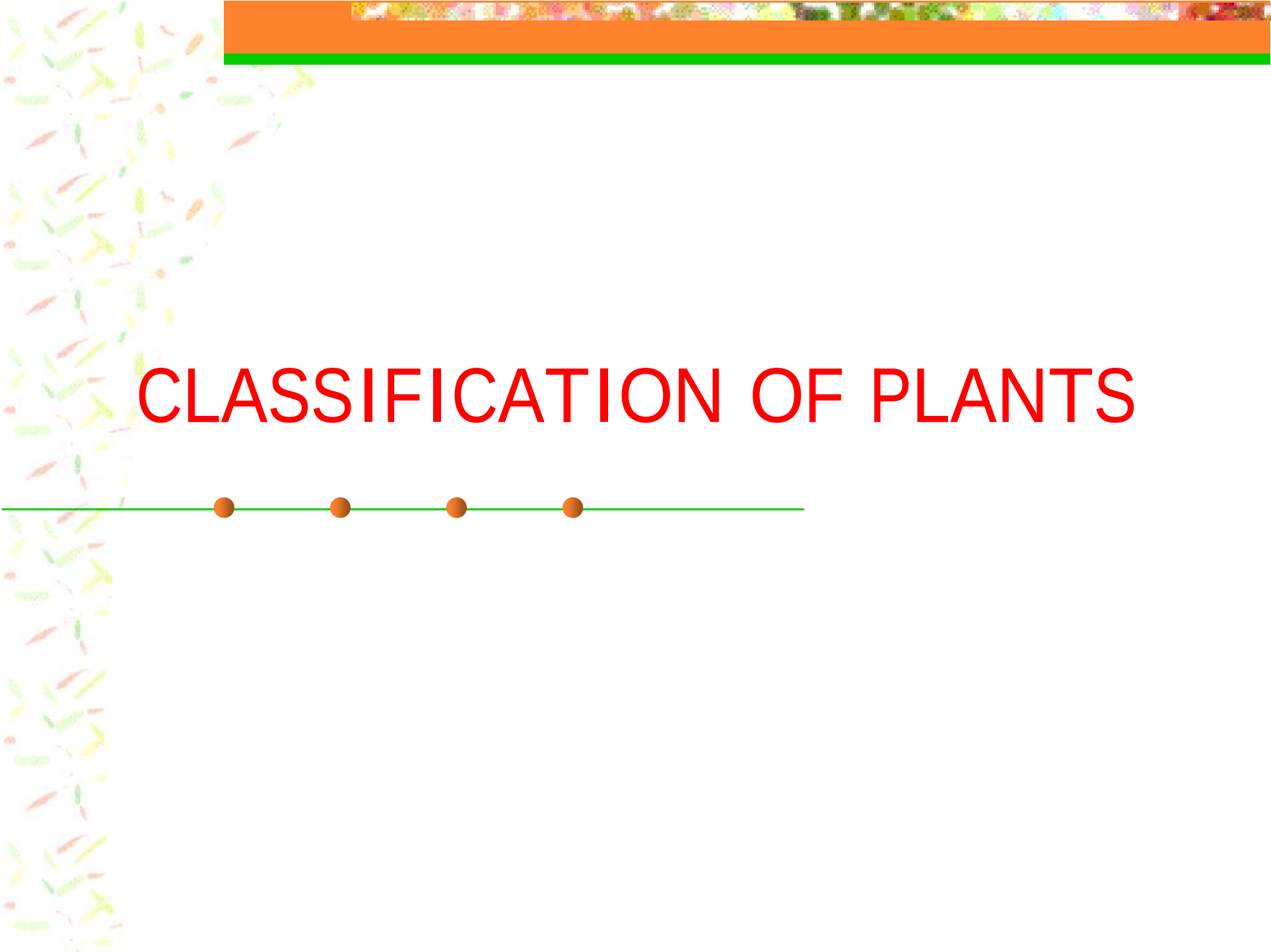
- 
- - **Medicine** – all prescription drugs are directly or derivatives of plants
 - - **Air** – Oxygen is a product of photosynthesis from plants
 - - **Habitat** – In addition to humans, all other organisms depend on plants to maintain the habitat
 - - **Climate** – plants store carbon and have helped in preserving CO₂ out of the atmosphere



BASIC CHARACTERS OF PLANTS

- 
- 1. They are multicellular in nature
 - 2. They are autotrophic (photosynthetic)
 - 3. They have eukaryotic cells
 - 4. They have cell walls made of cellulose

- 
- 5. They have structures adapted to survival out of water such as waxy cuticles on outer aerial surfaces and stomata on leaves
 - 6. Plants have a life cycle pattern which fluctuates between a haploid and diploid generations and sexual and asexual reproductions



CLASSIFICATION OF PLANTS



Introduction

- 
- Taxonomy is the science of classifying and identifying plants.
 - Scientific names are necessary because the same common name is used for different plants in different areas of the world.
 - Latin is the language used for scientific classification.

Karl von Linne (1707-1778)

- Swedish botanist
- Developed binomial classification scheme for plants.
- Uses two Latin words to indicate the genus and the species.
- Changed his name to the Latin name of Carolus Linnaeus.





Scientific Names

- 
- The first word is the genus and the second word is the species.
 - If there are additional words, they indicate the variety or cultivar.



Genus

- 
- Plants in the same genus have similar characteristics.
 - Examples:
 - Quercus – Oaks
 - Acer – Maples
 - Pinus – Pines
 - Ilex – Hollies
 - Cornus – Dogwoods
 - Ficus – Figs

Species

- Plants in the same species consistently produce plants of the same types.

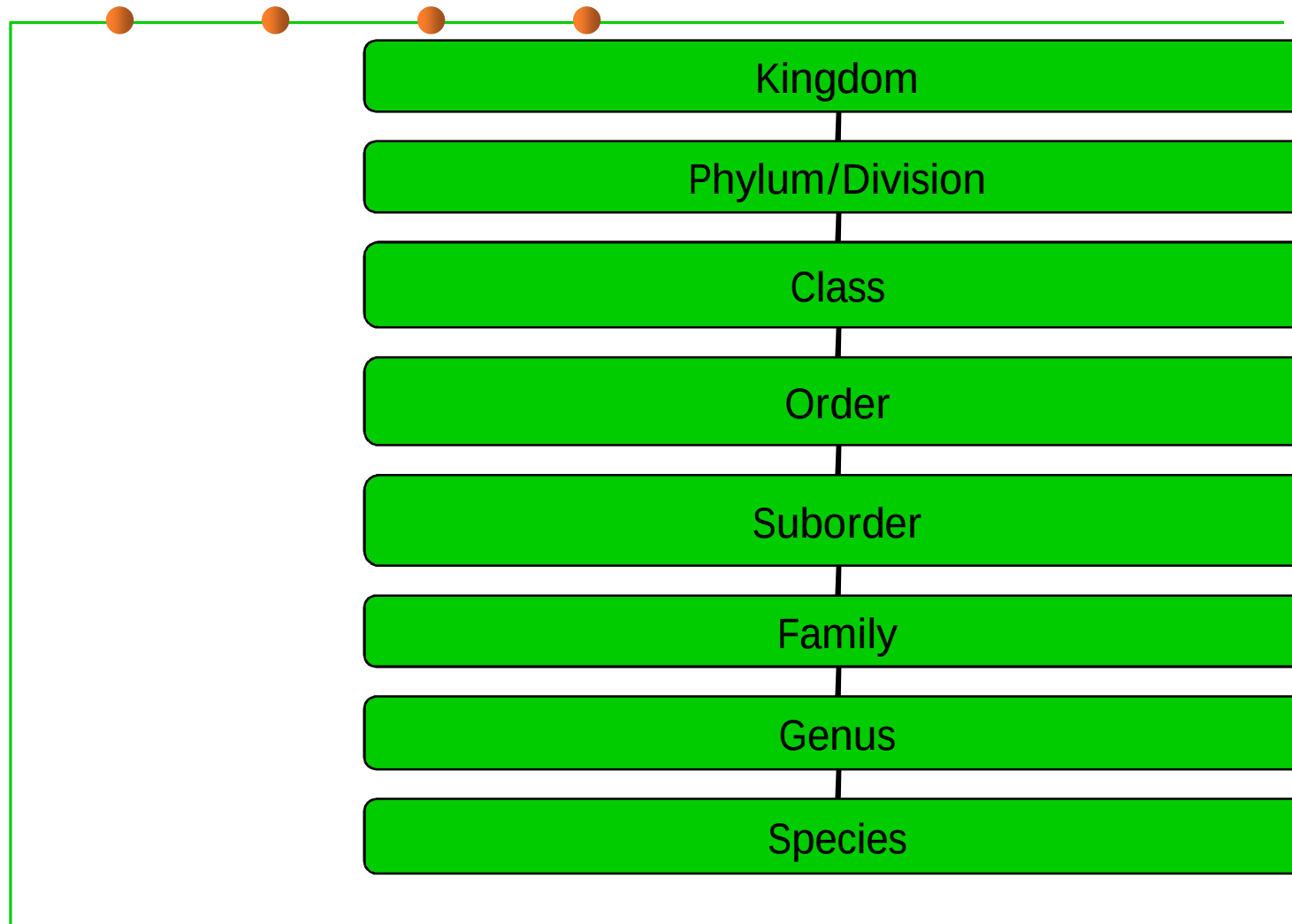




Scientific Classification

- 
- The broadest category of scientific classification is the Kingdom.
 - Either Plant or Animal
 - The broadest category of the plant kingdom is Division or Phylum.

Scientific Classification





Plant Kingdom

**Non-flowering
Plants**

**Flowering
Plants**



3 groups

Non - flowering Plants

Mosses

Ferns

Gymnosperms

Do NOT produce flowers



DIVISIONS

There are 12 divisions in the Plant Kingdom.

- **3** are **Bryophytes**
- **4** are **Seedless plants**
- **4** are **Gymnosperms** and
- **1** is **Angiosperms**.



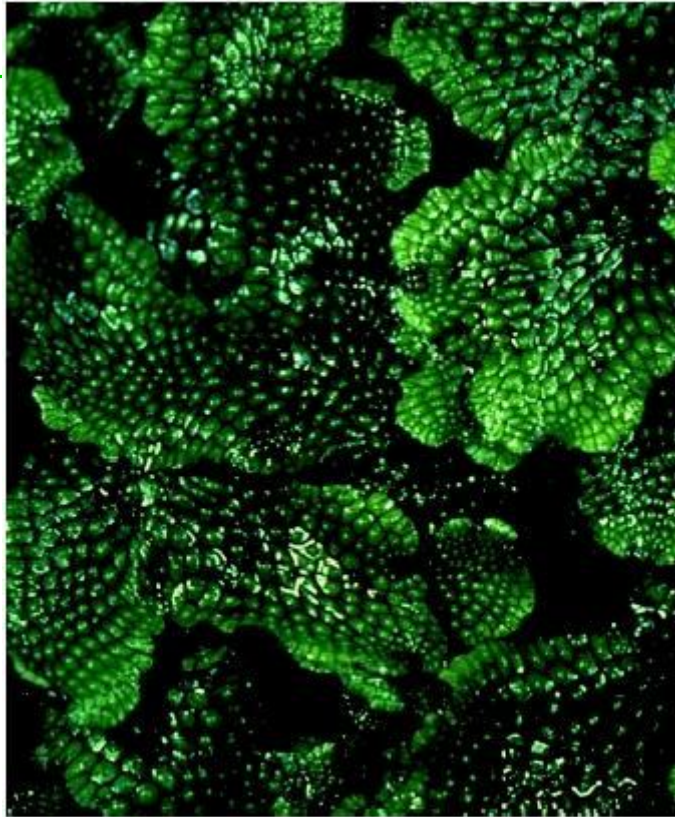
Divisions

- 
- The four most important divisions of the plant (Plantae) kingdom are:
 - Thallophytes: algae, fungi, and lichens
 - Bryophytes: mosses
 - Pteridophytes: ferns
 - Spermatophytes

Thallophytes



Bryophytes



Pteriophytes

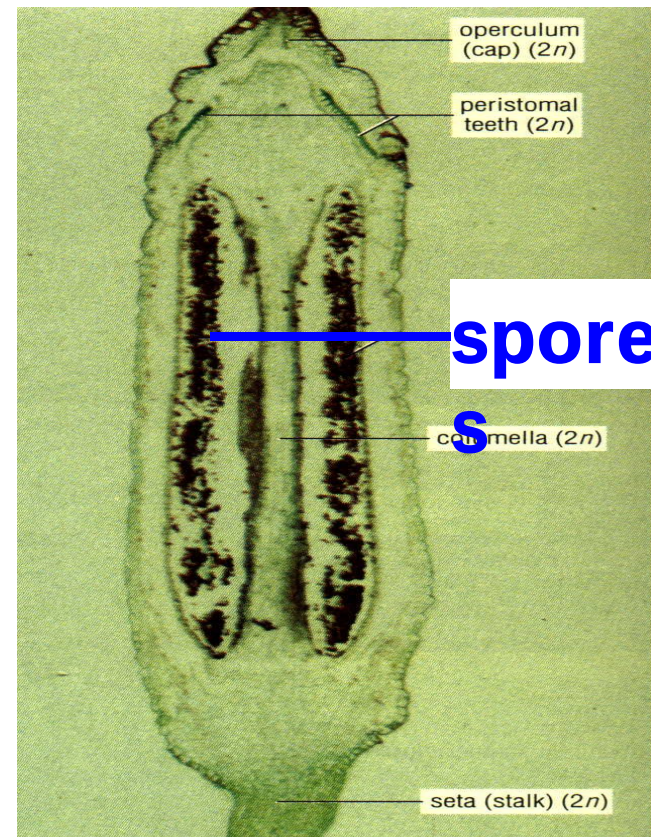
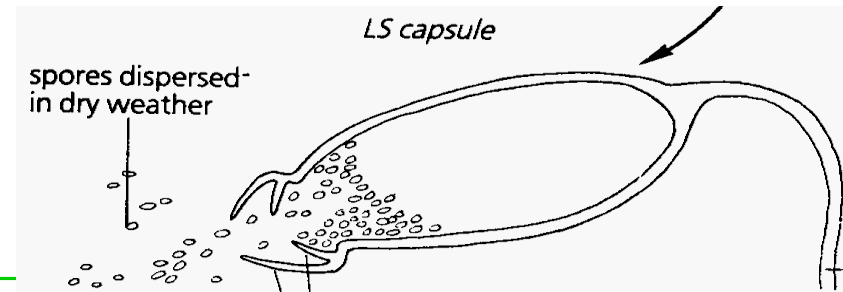


Examples of Mosses



Moss

Spore-producing capsule



Characteristics of Mosses

👉 Simplest plants

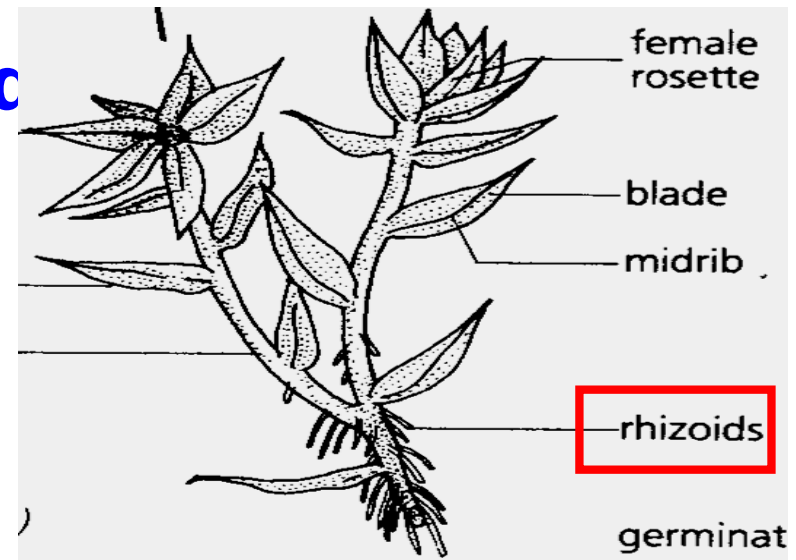
👉 No true roots, No vascular tissues (no trans

👉 Simple stems & leaves

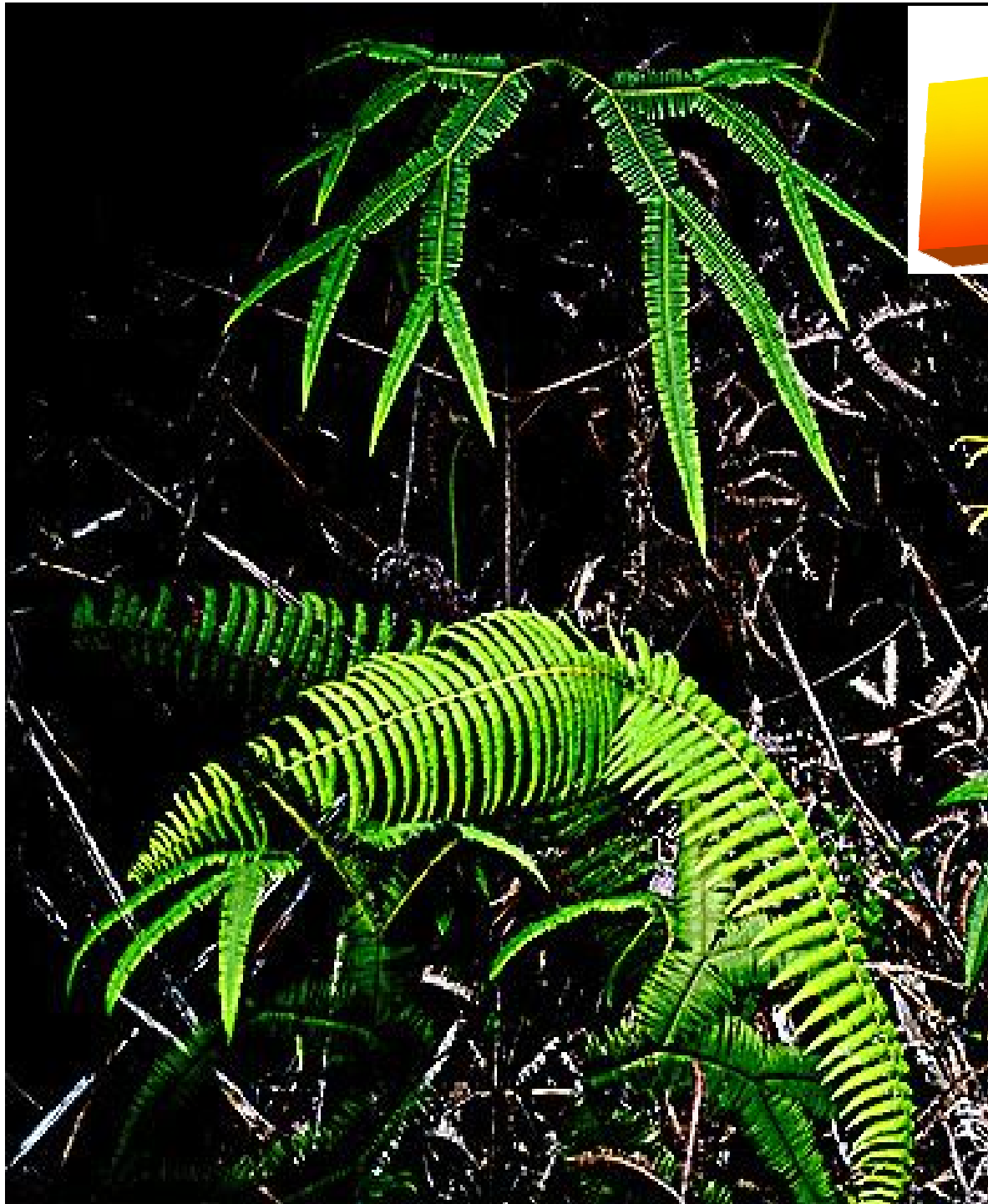
👉 Have **rhizoids** for anchorage

👉 **Spores** from capsules (wind-dispersal)

👉 **Damp** terrestrial land

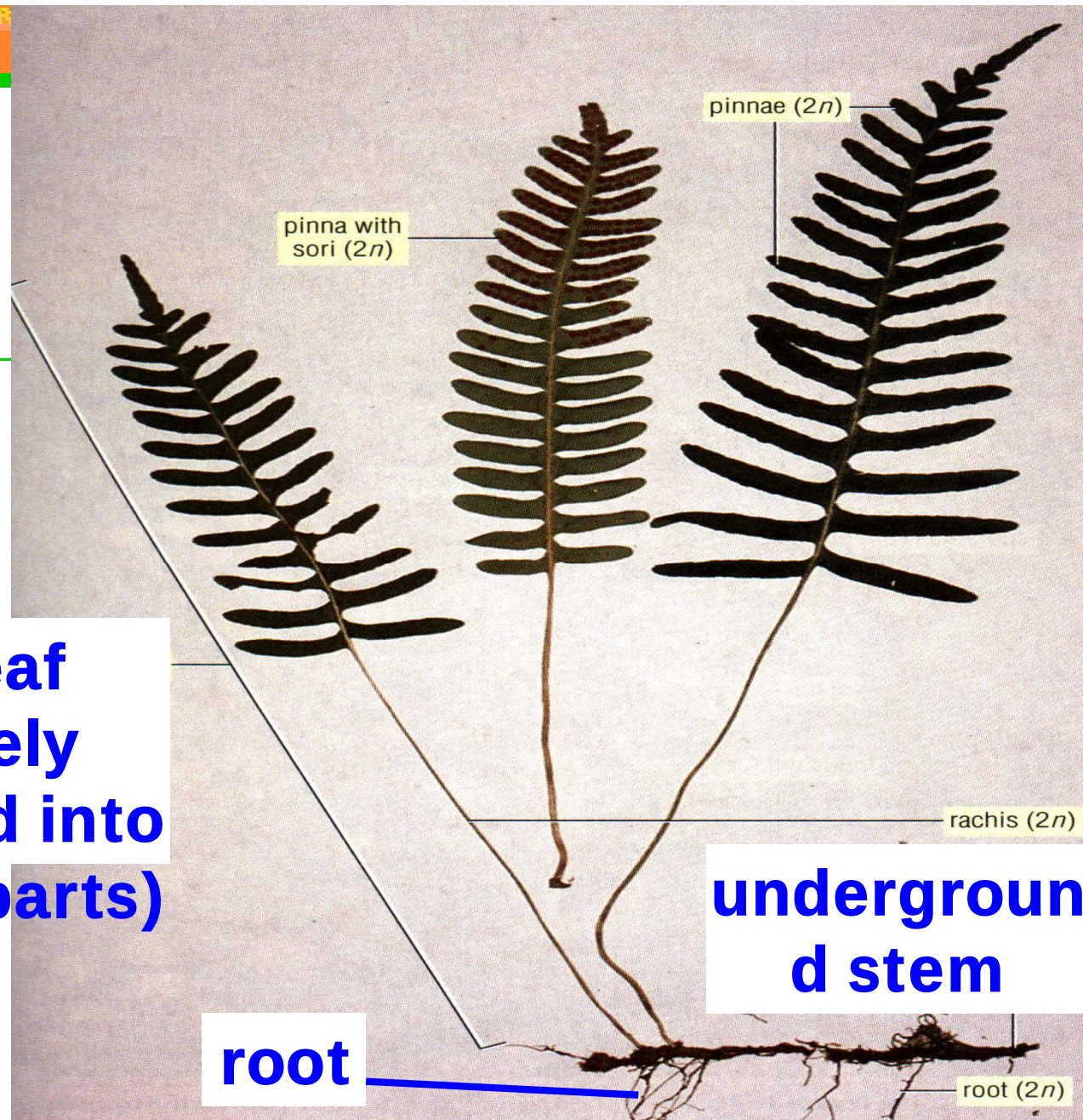


Fern



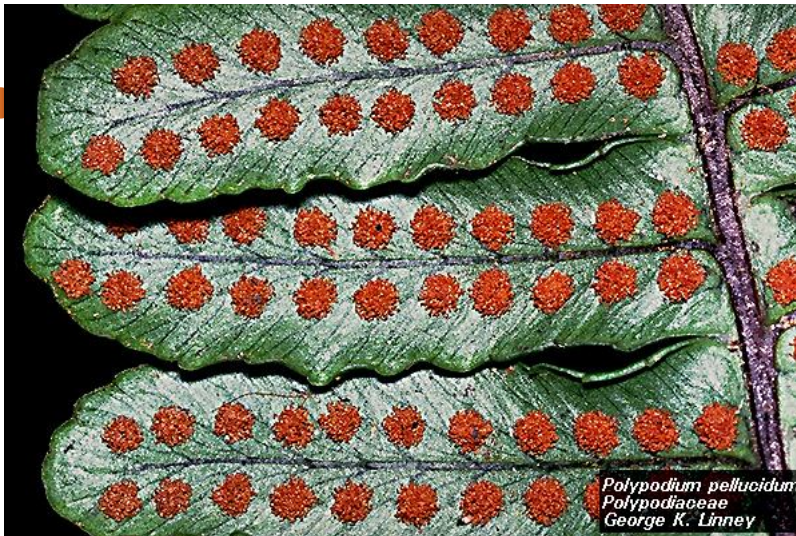
Fern

A leaf
(finely
divided into
small parts)



spore-producing
organs

(circinate)
young leaf



Characteristics of Ferns

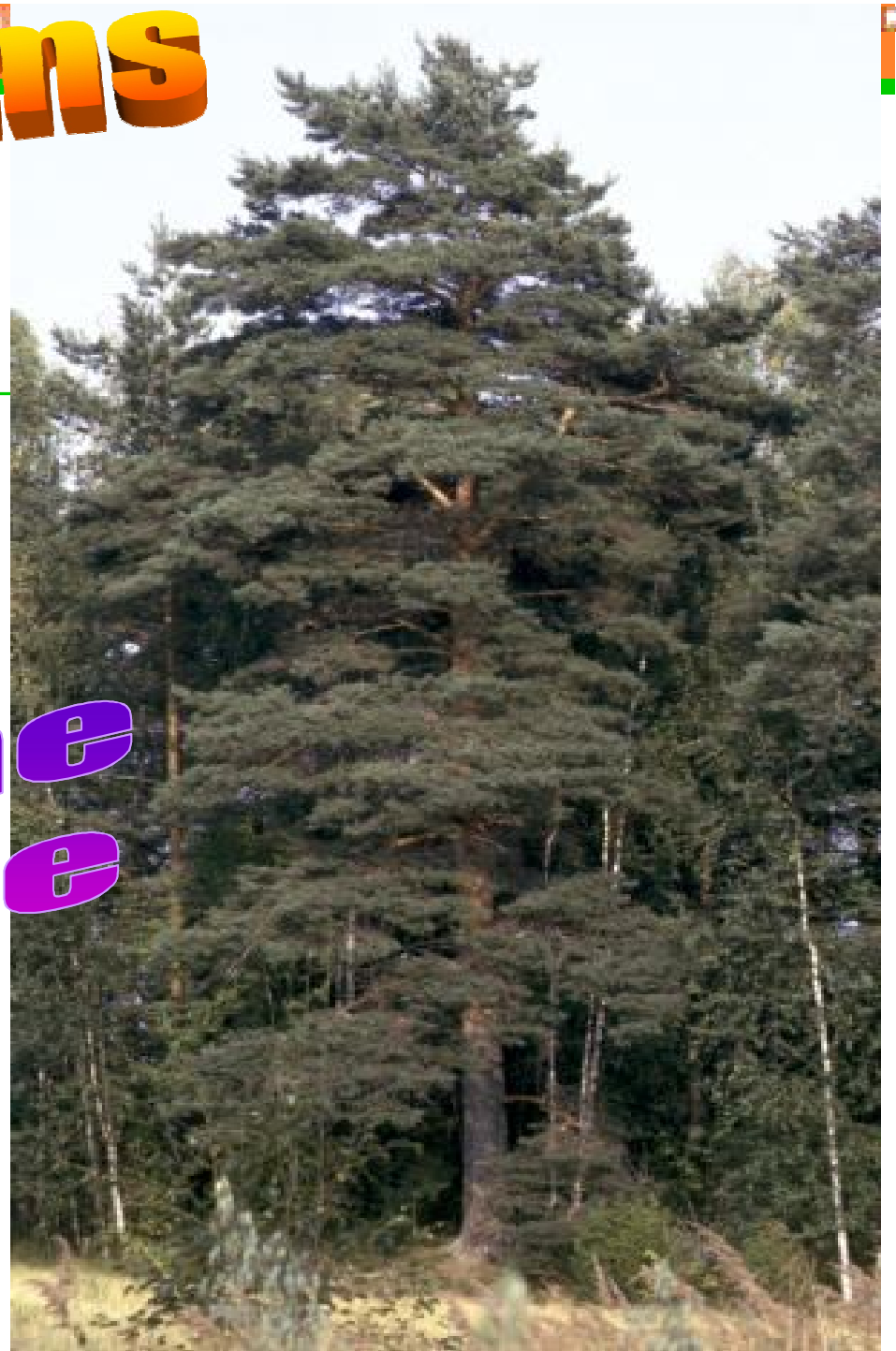
- ✍ roots, feathery leaves & underground stems
- ✍ have **vascular tissues** (transport & support)
- ✍ **Spore-producing organ** on the underside of leaves (reproduction)
- ✍ **Damp & shady** places



Gymnosperms



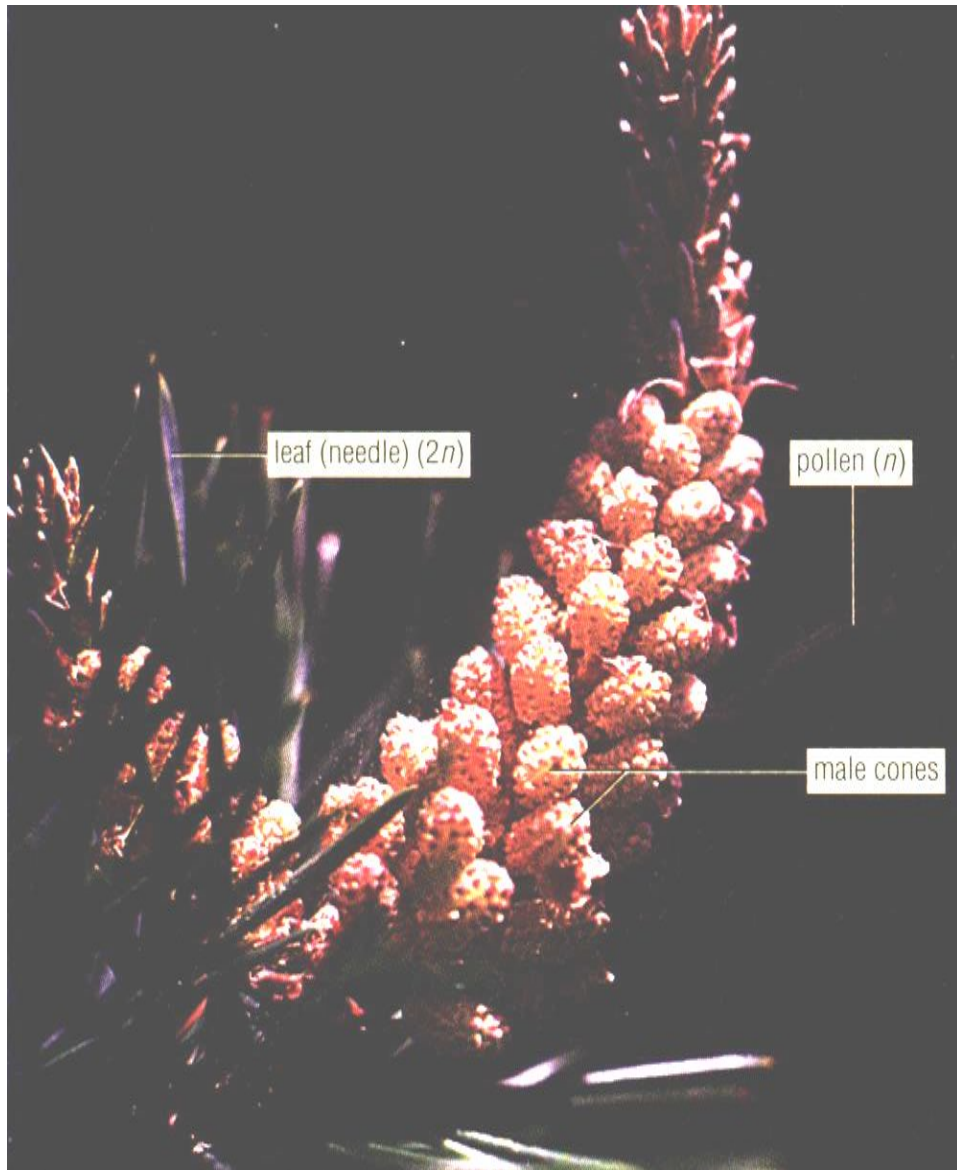
Pine
tree



needle-shaped leaves



Male cones (in clusters)



Female cones (scattered)



Characteristics of Gymnosperms

📌 tall evergreen trees

📌 roots, woody stems

📌 **needle-shaped** leaves

📌 **vascular tissues** (transport)

📌 **cones** with reproductive structures

📌 **naked seeds** in female cones

📌 dry places





2 groups

Flowering Plants



Monocotyledons Dicotyledons



roots, stems, leaves



vascular tissues (transport)



flowers, fruits (contain seeds)

Monocotyledon

s

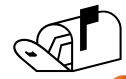


Parallel veins

Characteristics of Monocotyledons



one seed-leaf



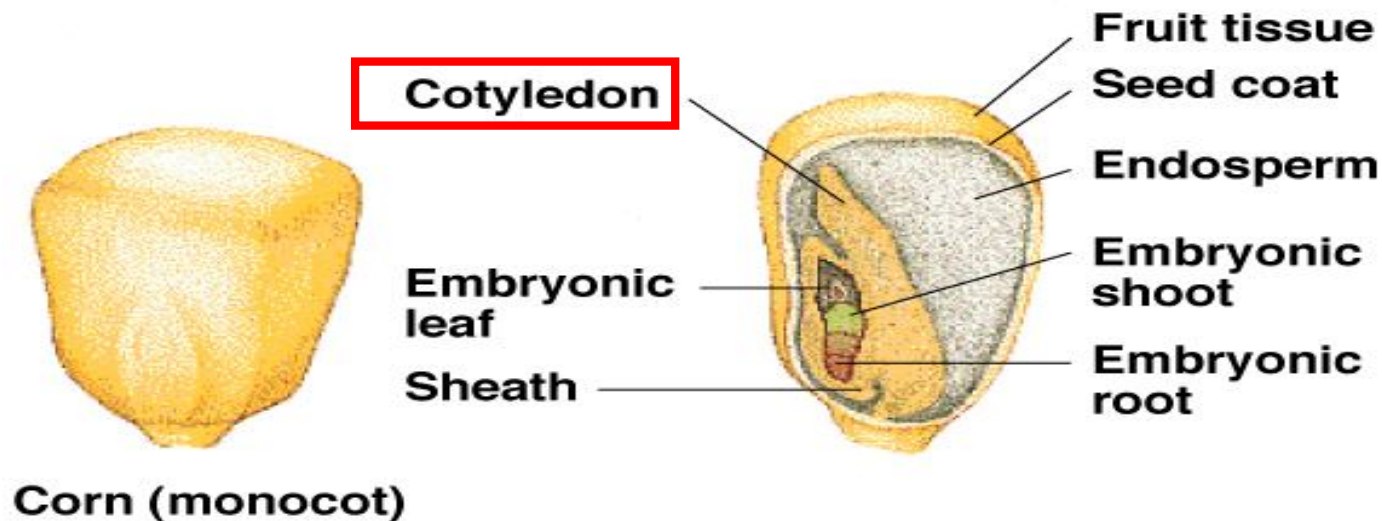
leaves have parallel veins



herbaceous plants



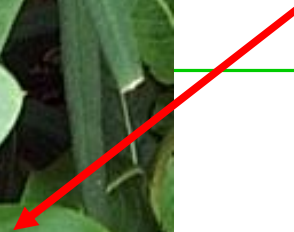
e.g. grass, maize



Dicotyledons



Veins in network



Characteristics of Dicotyledons



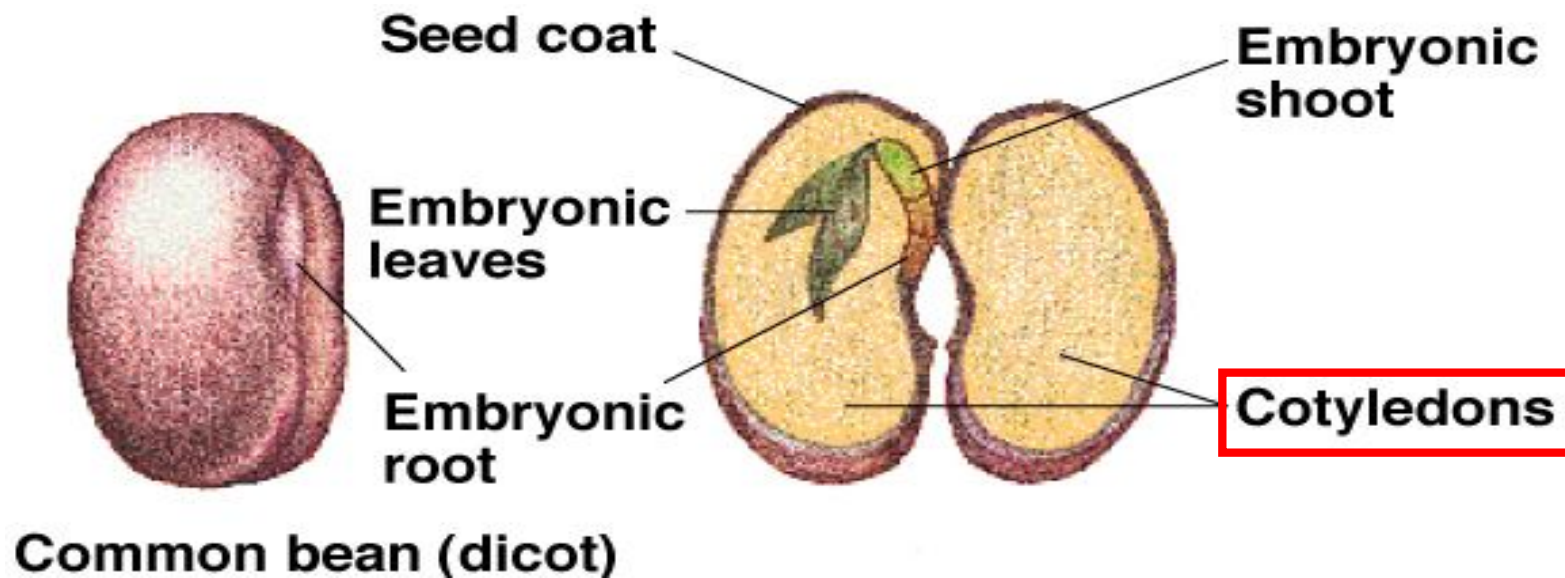
two seed-leaves



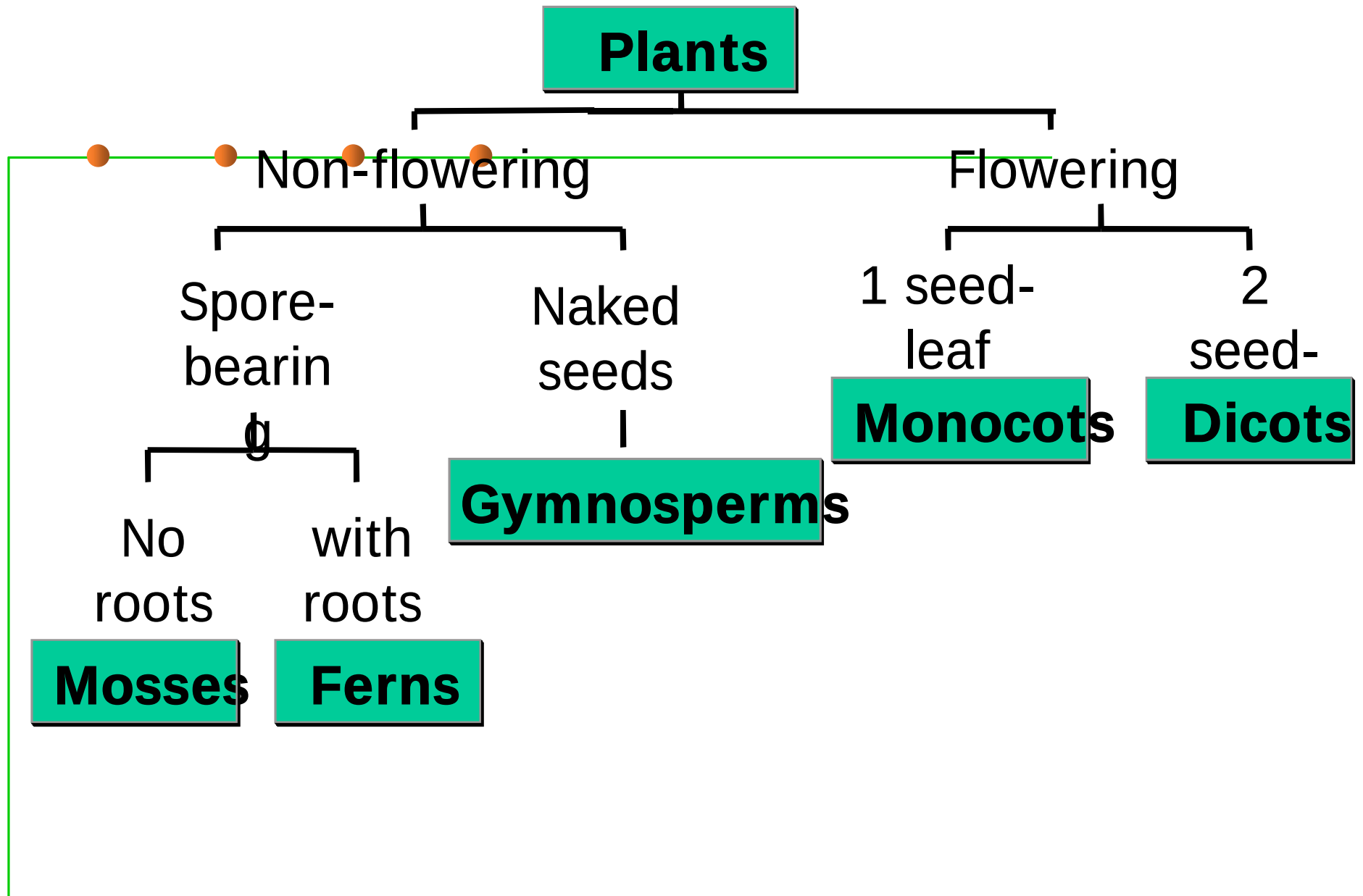
leaves have veins in
network



e.g. trees, sunflower, rose



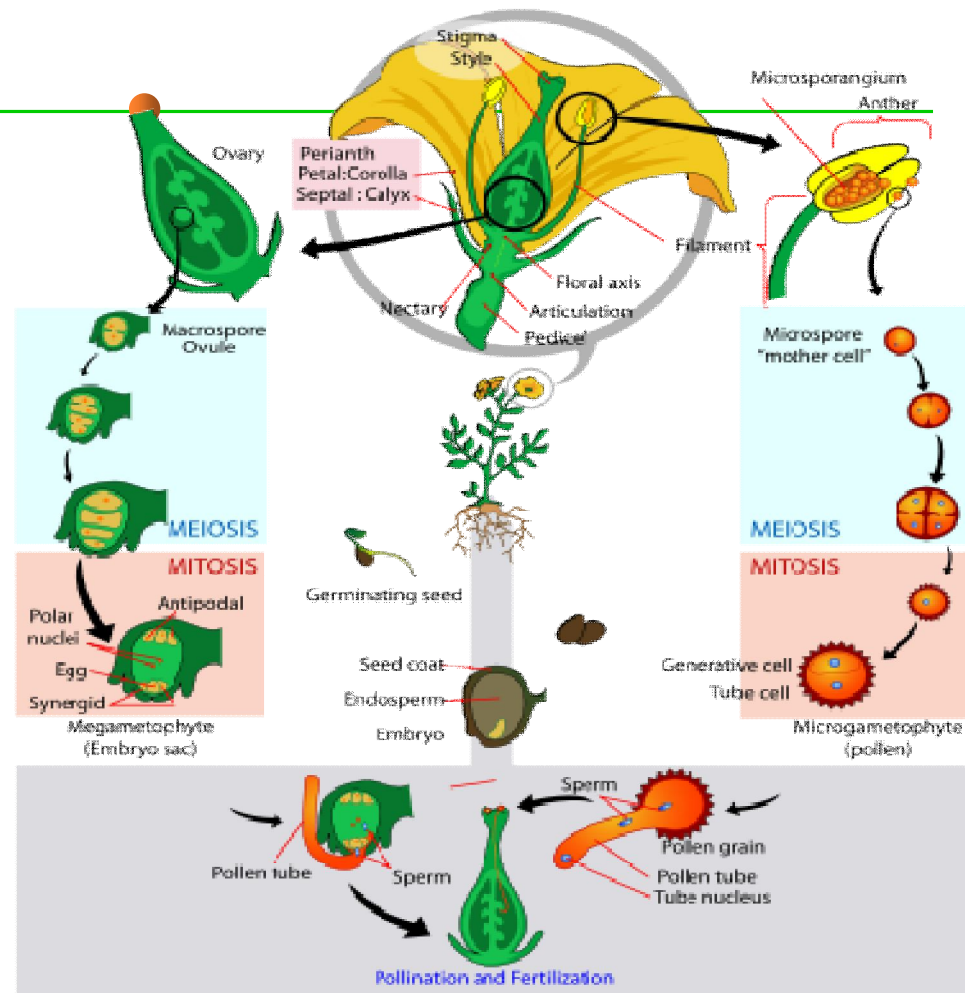
Plant Classification



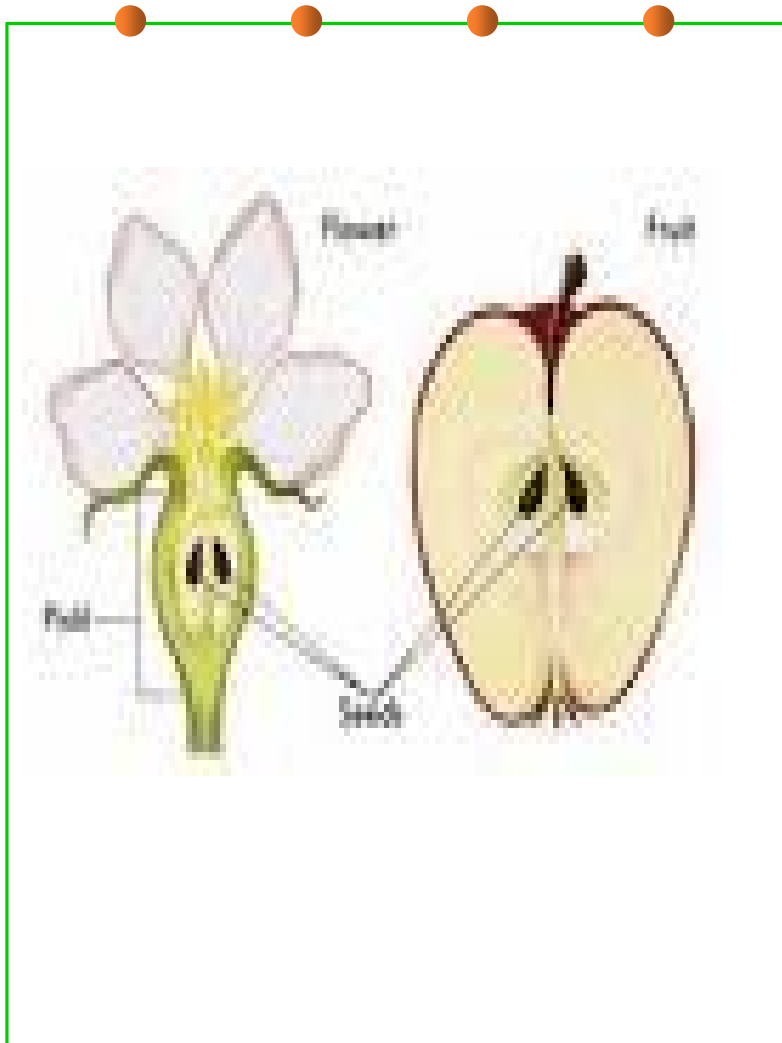
Gymnosperms



Angiosperms



Angiosperms



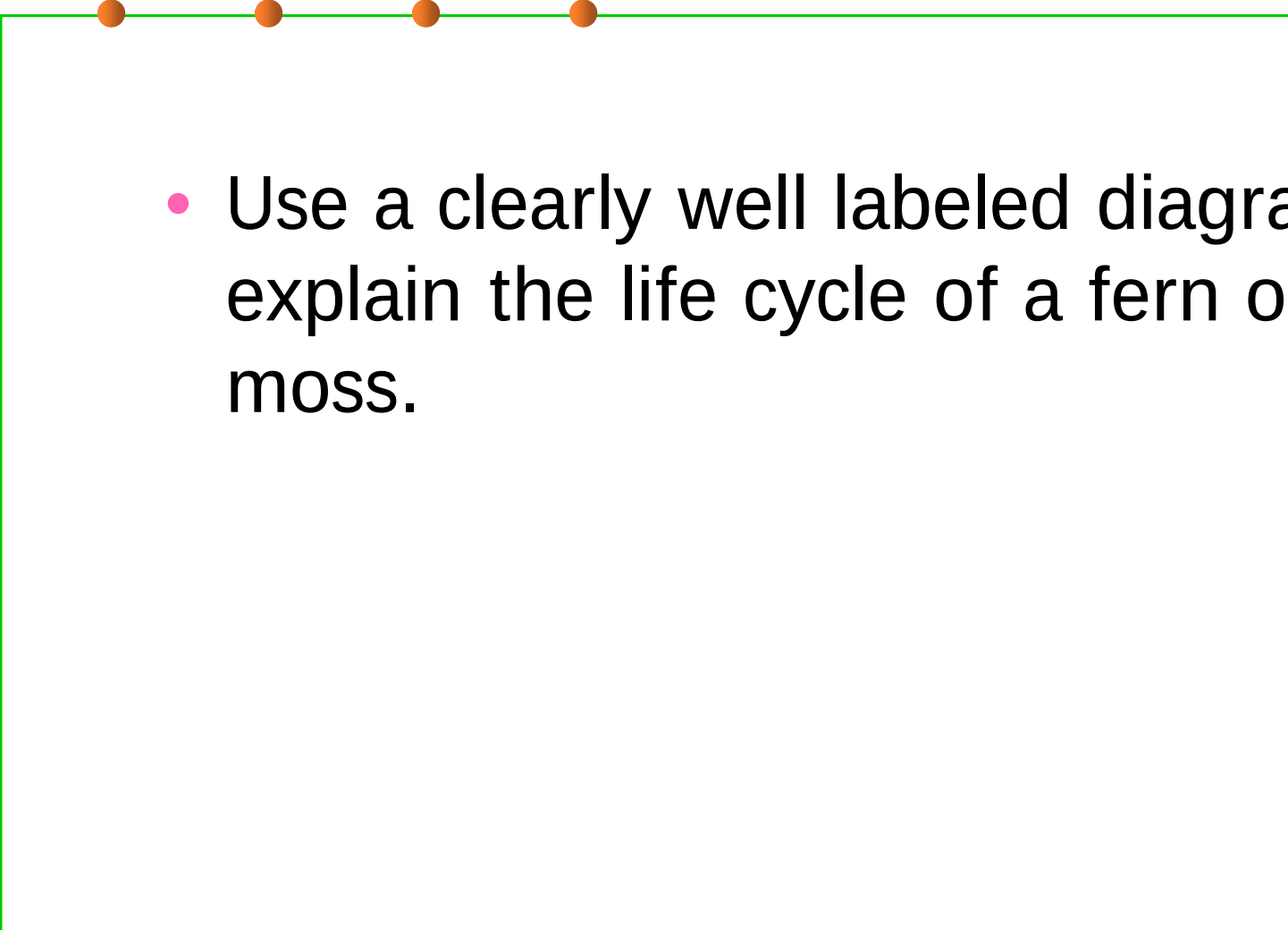


Common Plant Genus

- 
- Pinus-Pine
 - Acer-Maple
 - Ilex-Holly
 - Ficus-fig
 - Cornus-dogwood
 - Rhododendron-rhododendron
 - Quercus-oak



Take-Home CA Assignment:

- 
- Use a clearly well labeled diagram to explain the life cycle of a fern or moss.



Plant Uniqueness



Identifying Plants

- 
- Physical characteristics are used to identify plants which include....
 - Life Cycle
 - Form
 - Foliage Retention
 - Plant Parts
 - Use & Location

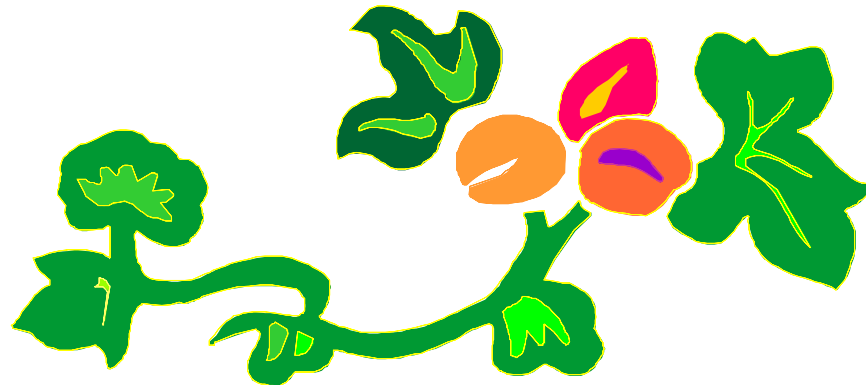


Life Cycle

- 
- Annuals
 - Plants that complete their life cycle in one year.
 - Biennials
 - Plants that complete their life cycle in two years.
 - Perennials
 - Plants that live more than two years.

Growth Habits

- Trees
- Shrubs
- Vines





Growth Forms

- 
- Columnar
 - Spreading
 - Weeping

- Round
- Oval
- Pyramidal

Growth Forms



Growth Forms





Foliage Retention

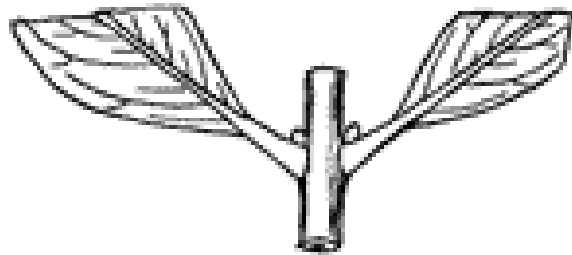
- 
- Deciduous
 - Loses leaves during the dormant season.
 - Evergreen
 - Keeps leaves and remains green year-round.



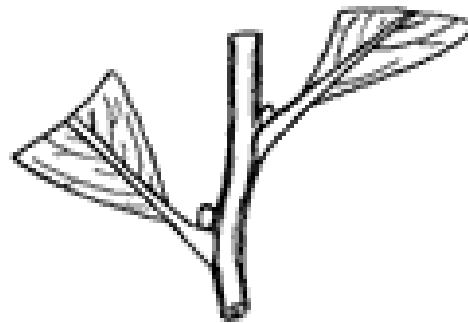
Plant Parts – Leaf

- 
- Arrangement
 - Shapes
 - Color
 - Vein Pattern
 - Form – Simple or Compound
 - Margin
 - Surface

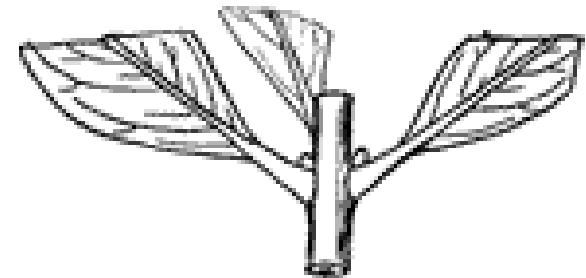
Leaf Arrangement – Simple



opposite

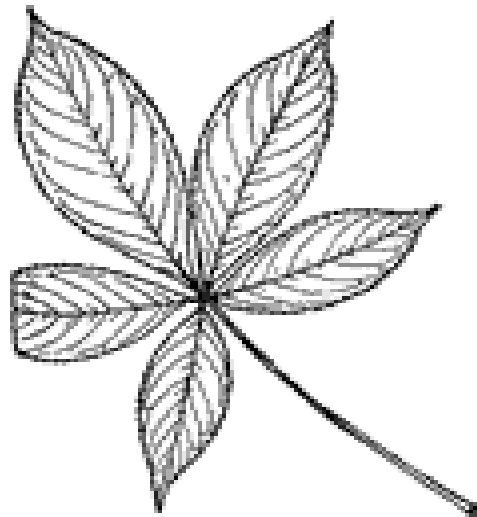


alternate

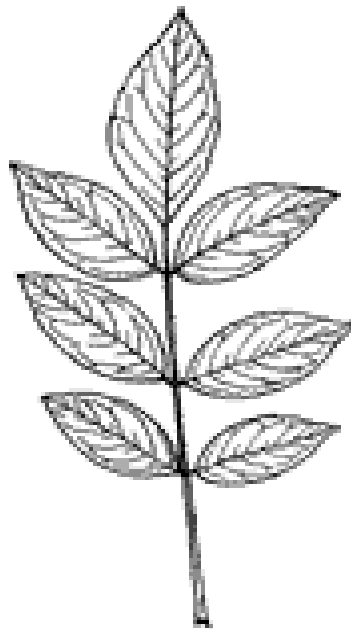


whorled

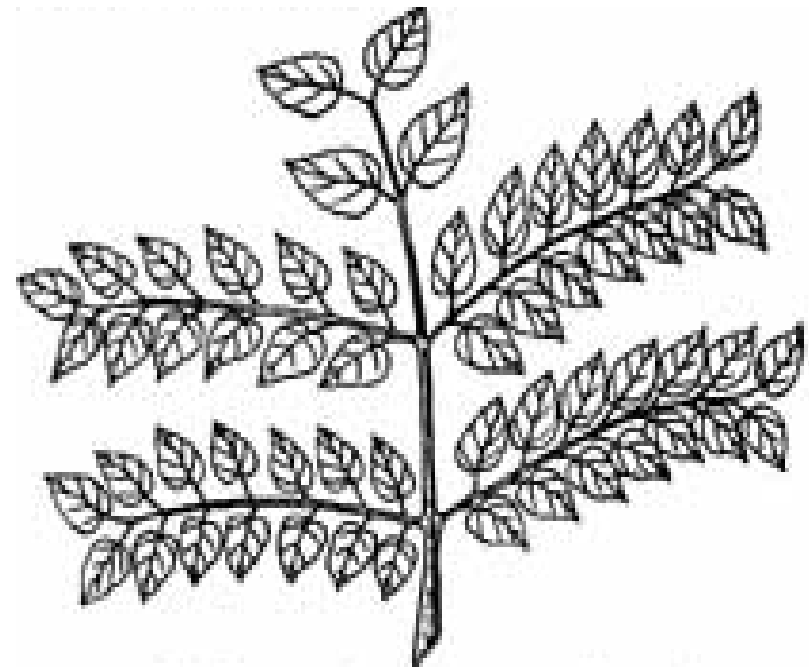
Leaf Arrangement – Compound



palmately
compound

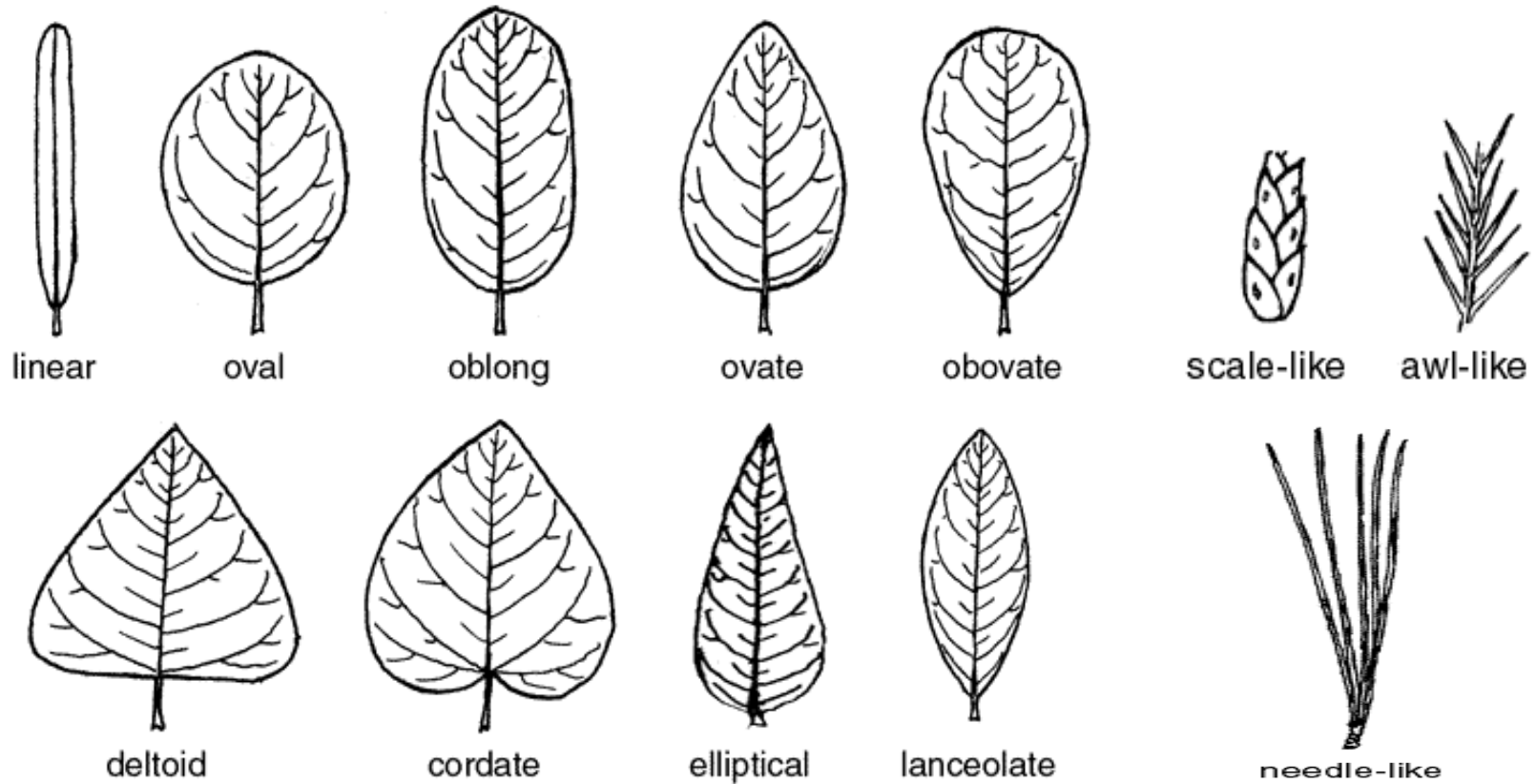


pinnately
compound



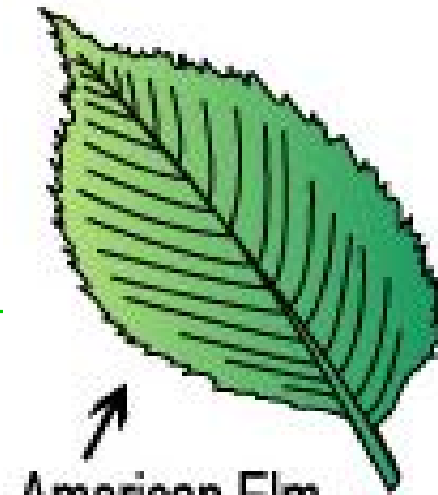
Bi-Pinnately Compound Leaf

Leaf Shape



Vein Pattern

- Pinnate
- Palmate
- Parallel
- Dichotomous



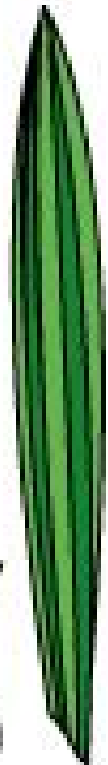
American Elm
(Pinnate)



Ginkgo
(Dichotomous)



Sweetgum
(Palmate)



Corn
(Parallel)

Leaf Margin



entire



undulate



finely
serrate



coarsely
serrate



doubly
serrate



crenate



lobed



Leaf Surface

- There are 8 common leaf surfaces.
 - Glabrous
 - Pubescent
 - Villous
 - Tomentose
 - Scabrous
 - Glaucous
 - Rugose
 - Glandular

Leaf Surface – Glabrous

- The surface is smooth, not hairy.



Leaf Surface – Pubescent

- Short, soft hairs cover the surface.



Leaf Surface – Villous

- Long, straight hairs cover the surface.



Leaf Surface – Tomentose

- Covered with wool-like hair.



Leaf Surface – Scabrous

- Covered with short, prickly hairs.



Leaf Surface – Glaucous

- Covered with a bluish-white waxy substance.



Leaf Surface – Rugose

- Surface is wrinkly.



Leaf Surface – Glandular

- Glands filled with oil or resin cover the surface.



Prunus tomentosa

Plant Parts – Flowers

- Color
- Shape
- Size



Plant Parts – Bud & Stem

- Shape & Color
- Stem Modifications
 - Thorns
 - Spines
 - Prickles



Plant Parts – Modified Stems



Plant Parts – Roots

- Tap
- Fibrous
- Bulb



Plant Parts – Roots



Tap Root



Tuberous Root



Fibrous Root

Plant Parts – Fruit

- Cones (Pines, Spruces, Firs)
- Nuts (Chestnuts, Peanuts, Cashew nuts)
- Pomes (Apple, Pear)
- Drupes (Mango, Peach, Date, oilpalm, coconut)
- Brambles (Blackberry bush, Raspberries)
- Capsules (Baobab, Cotton, Willow)
- Samara (Combretum, Maple)



Plant Parts – Fruit



Plant Parts – Fruit





Use & Location

- 
- Not absolute, but helpful.
 - Indoor or outdoor.
 - Altitude
 - Wet or dry
 - Hardiness Zone
 - Sun, partial shade, or shade.
 - Landscape purpose – specimen, border, etc.



Factors that facilitated the adaptation of plants to life on land

- Possession of refined vascular systems.
Compare water conducting cells in bryophytes, tracheids in gymnosperms and vessels angiosperms
- Transition from water dependent reproduction to water independent reproduction
- Possession of stomata, cuticle and pores which enabled plants to control loss of water in dry environments

- 
- 
- Evolution of lignin and woody skeletons
 - Elaborate roots and leaves for photosynthesis to manufacture food



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